

A BEGINNER-ACCESSIBLE QUANTITATIVE STUDY OF

MEAN-REVERSION TRADING IN THE CRUDE OIL MARKET

*A Backtest of a Simple Rule-Based Strategy on 11.5 Years of Real Market Data (2015–2026),
with Complete Explanations of Every Concept, Term, and Calculation*

Data sources: Yahoo Finance (daily futures prices) · U.S. Energy Information Administration (weekly inventory and refinery statistics)

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Abstract

This report documents, from first principles, the design and evaluation of a simple rule-based trading strategy for crude oil, the world's most heavily traded commodity. The strategy tested is a "mean-reversion" rule: buy West Texas Intermediate (WTI) crude oil whenever its price falls more than 5% below its own 20-day average and sell as soon as the price recovers back above that average. The rule was evaluated by "back testing" (replaying it, day by day) against 2896 days of real historical prices spanning 2015-01-02 to 2026-07-10 (11.5 years), a period that includes the 2020 COVID-19 price collapse and the 2022 energy crisis.

Over the full period the rule generated 59 completed trades, of which 42 were profitable (a win rate of 71.2%). The sum of all trade returns was +74.5 percentage points, compared with +35.5% for simply buying oil at the start and holding - an outperformance of roughly 39 percentage points, before trading costs. However, the study also exposes a serious weakness: because the rule contains no protective exit ("stop-loss"), its single worst trade lost 39.8% during the 2020 crash, and the average losing trade (-8.76%) was substantially larger than the average winning trade (+5.32%). The report explains every term, dataset, formula, and chart used, so that a reader with no prior knowledge of finance or commodities can follow and reproduce the entire analysis in an ordinary spreadsheet.

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1. Introduction

1.1 What this report is about

Commodities are basic physical goods that are bought and sold in bulk: crude oil, natural gas, copper, wheat, coffee. Unlike a company share, a barrel of oil is the same product whoever produces it, so its price is set almost entirely by global supply and demand. "Commodity trading" is the business of buying and selling these goods or financial contracts linked to them to earn a profit from changes in their prices.

This report asks a very concrete question: can an extremely simple, mechanical trading rule (one with no human judgement at all) have made money in the crude oil market over the last decade? To answer it, we use a standard research technique called a backtest: we write the rule down precisely, then replay history day by day and record every trade the rule would have made. Because the rule is applied to data it has never "seen," a backtest is the closest thing finance has to a laboratory experiment.

1.2 Why crude oil

Crude oil is unrefined petroleum (the dark liquid extracted from underground reservoirs before it is processed into petrol, diesel, jet fuel, and plastics.) It is measured in barrels (abbreviated "bbl"), where one barrel equals 159 litres. Roughly 100 million barrels are consumed worldwide every day, making oil the largest commodity market on Earth. Its price affects the cost of nearly everything that is grown, manufactured, or transported, which is why oil prices appear in the news almost daily and why the major commodity trading houses (Vitol, Trafigura, Glencore, Mercuria, Gunvor) all trade it in enormous volume.

1.3 Objectives

- **Objective 1:** Define and explain every concept needed to understand oil markets at an introductory level;
- **Objective 2:** Build a daily market-tracking dataset from authoritative public sources;
- **Objective 3:** Specify a mean-reversion trading rule precisely enough that a spreadsheet can execute it;
- **Objective 4:** Backtest the rule on 11.5 years of real prices and report standard performance statistics;
- **Objective 5:** Assess honestly what the results do and do not prove.

1.4 How to read this report

No prior knowledge is assumed. Section 2 builds the vocabulary: what WTI and Brent are, what a futures contract is, what a moving average is, and what the supply-side statistics (inventories, refinery utilisation) mean. Section 3 describes the data and the method step by step, exactly as implemented in the accompanying Excel workbook. Section 4 presents the results with figures and tables, each one explained in plain language. Section 5 discusses limitations, and Section 7 is a full alphabetical glossary you can consult whenever a term is unclear.

2. Background: The Concepts, Explained From Zero

2.1 Crude oil and the barrel

Crude oil straight from the ground varies in quality. Two properties matter most for its price. Density ("light" vs "heavy"): light crude flows easily and yields more petrol when refined, so it is worth more. Sulphur content ("sweet" vs "sour"): sweet crude (low sulphur) is cheaper to refine into clean fuels, so it also commands a premium. Prices are always quoted in US dollars per barrel (\$/bbl).

2.2 WTI and Brent: the two reference prices

Because thousands of slightly different crude oils exist, the market prices them relative to a small number of standard reference grades called benchmarks. The two that dominate the news are:

- **WTI** (West Texas Intermediate): A light, sweet crude produced in the United States and delivered at a huge tank farm in Cushing, Oklahoma. WTI is the main US price benchmark and trades on the New York Mercantile Exchange under the ticker symbol CL. In this report, data labelled “CL=F” is the WTI futures price as published by Yahoo Finance.
- **Brent**: A light, sweet crude from the North Sea (between the UK and Norway). Brent is the benchmark used to price roughly two-thirds of the world's internationally traded oil, and trades on the ICE exchange in London (ticker BZ; “BZ=F” on Yahoo Finance).

The difference between the two prices (the Brent–WTI spread) is itself informative. Brent normally trades a few dollars above WTI, mainly because moving US crude to the global market costs money (pipelines, ships). When the spread widens sharply, it usually signals a local imbalance, such as too much oil accumulating inside the United States.

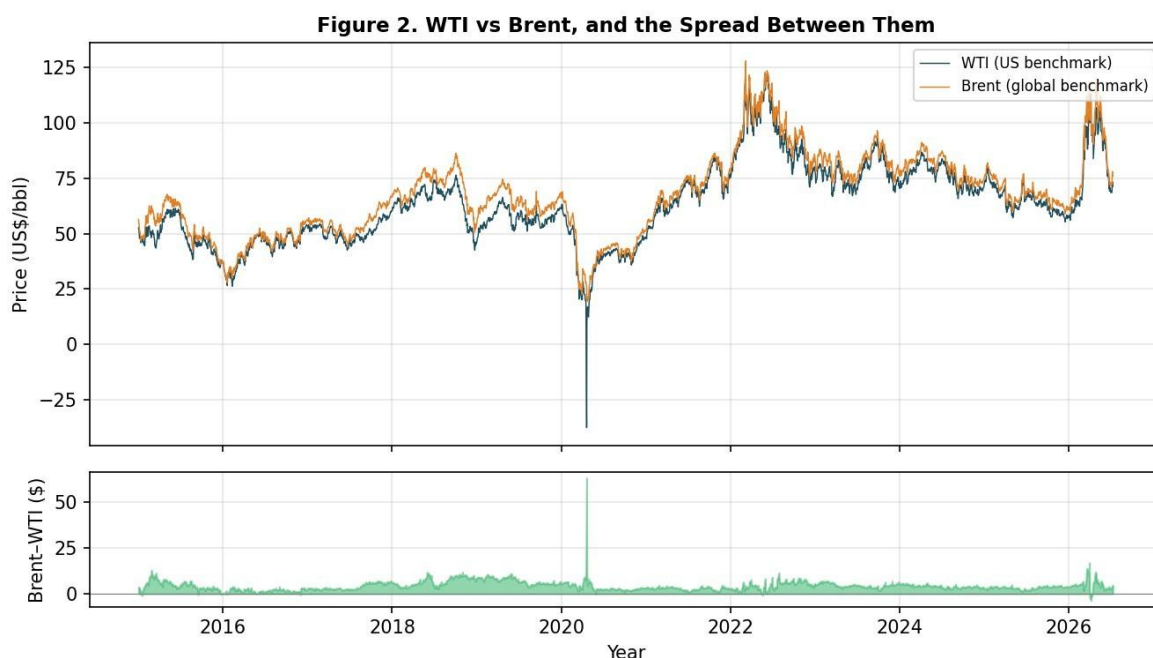


Figure 2. Top: daily closing prices of WTI (dark blue) and Brent (orange), 2015–2026. Bottom: the spread (Brent price minus WTI price). Note that the two lines move almost together (they are the same underlying commodity) while the spread (green) fluctuates within a few dollars.

2.3 Futures contracts: how oil is traded

Almost nobody who trades oil handles physical barrels. Instead, they trade futures contracts: standardised agreements to buy or sell 1,000 barrels of a specific crude at a fixed price on a fixed future date. Futures exist so that producers and consumers can lock in prices in advance (this protective use is called hedging), but they also allow traders to profit from price movements without ever touching a barrel: a trader who buys a futures contract at \$70 and sells it back at \$75 earns \$5 per barrel, i.e. \$5,000 per contract.

The "front-month" contract is the one expiring soonest, and its price is what news reports mean by "the price of oil." All prices used in this study are front-month futures closing prices (the last traded price of each day.)

One more concept is worth knowing because it appears constantly in commodity markets: the shape of the futures curve. If you list the prices of contracts expiring 1, 2, 3, ... 12 months from now, they form a curve. When later months are more expensive than earlier ones the market is in contango, which typically signals comfortable supply (oil is abundant today, so today's price is depressed relative to the future). When later months are cheaper the market is in backwardation, signalling tight supply (buyers pay a premium to get oil now). Figure 4 illustrates both shapes schematically.

Figure 4. The Two Shapes of the Futures Curve

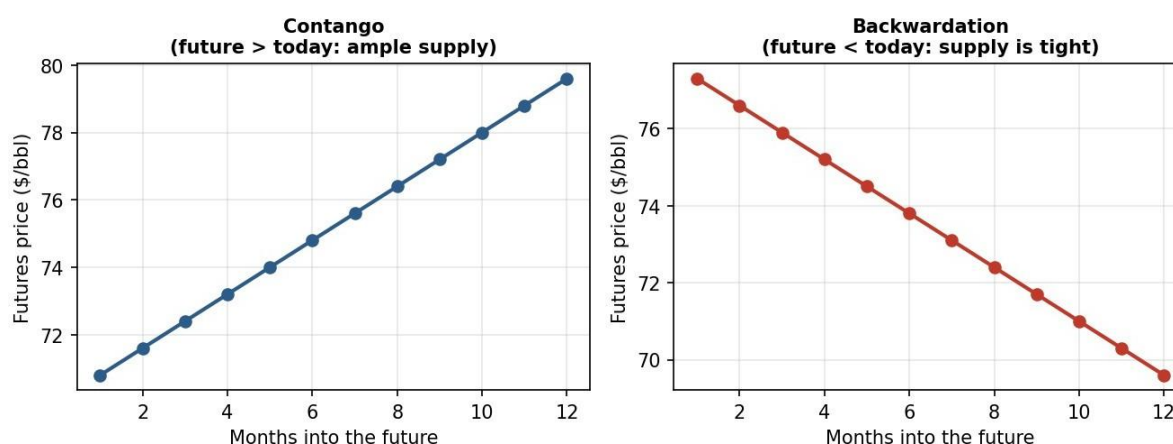


Figure 4. Schematic illustration (not real data) of the two possible shapes of a futures curve. Left: contango (each later delivery month costs more.) Right: backwardation (each later month costs less, indicating urgent demand for oil today.)

2.4 The supply-side statistics: inventories and refinery utilisation

Price charts tell you what happened; supply statistics help explain why. Two weekly numbers published every week by the EIA (the U.S. Energy Information Administration, the statistics agency of the US Department of Energy) are watched by every oil trader in the world:

- **US crude inventory:** The total amount of crude oil sitting in storage tanks in the United States (excluding the government's strategic reserve), measured in millions of barrels. Rising inventories mean supply is outpacing demand (bearish, prices tend to fall); falling inventories mean demand is outpacing supply (bullish, prices tend to rise).
- **Refinery utilisation:** The percentage of total US oil-refining capacity actually running that week. Refineries are the customers of crude oil (they buy crude and turn it into petrol and diesel). High utilisation (say 95%) means refineries are buying crude aggressively; low utilisation (below 85%, e.g. during maintenance season or a recession) means crude demand is weak.

These two series are included in the accompanying workbook's Tracking sheet and are analysed for the most recent 12 weeks in Section 4.4.

2.5 The moving average: smoothing out the noise

Daily prices are noisy: they jump up and down for countless small reasons. A moving average smooths this noise. The 20-day moving average (written 20-day MA, or MA20) on any given day is simply the arithmetic mean of the last 20 daily closing prices. Twenty trading days is roughly one calendar month, so the MA20 answers the question: "what has oil cost, on average, over the past month?" When today's price is far below its MA20, the market has fallen hard and fast relative to its own recent history.

2.6 Mean reversion: the hypothesis being tested

Mean reversion is the tendency of a price that has moved sharply away from its recent average to drift back toward it. The economic intuition in oil is physical: if prices crash 5–10% in a few days, refiners find crude suddenly cheap and buy more, some high-cost producers cut output, and bargain-hunting traders step in, all of which pushes the price back up. The opposite forces cap sharp rallies. Mean reversion is a hypothesis, not a law; the entire purpose of this study is to test whether it held in oil, and when it failed.

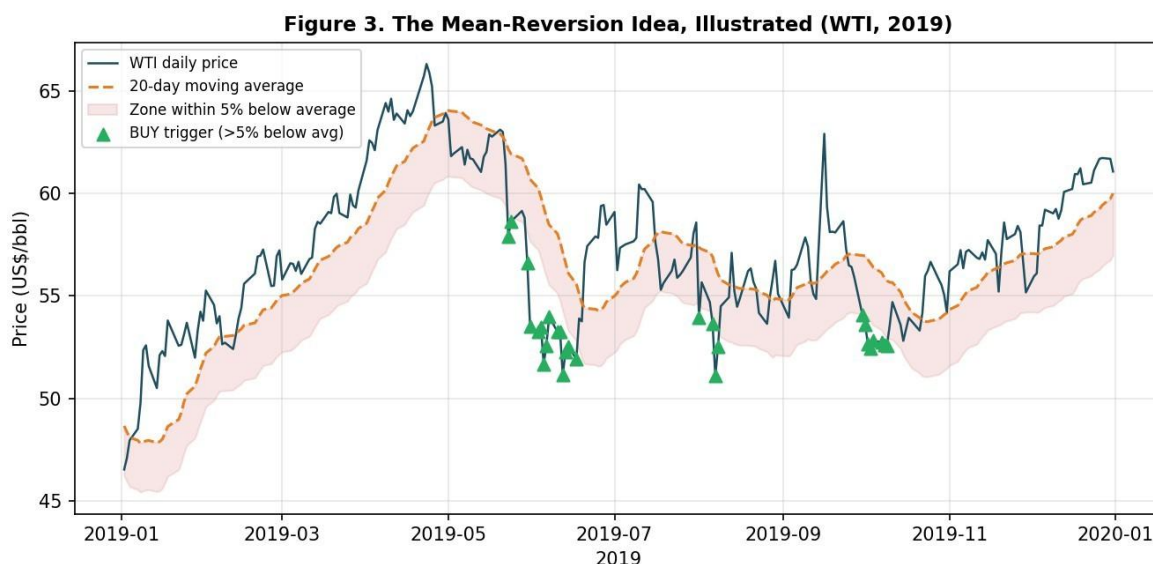


Figure 3. The idea illustrated on one year of real data (WTI, 2019). The solid line is the daily price; the dashed orange line is its 20-day moving average; the pink band marks the zone up to 5% below the average. Green triangles mark days when the price closed more than 5% below its average — the strategy's BUY trigger. Note how, in this year, prices repeatedly fell through the band and then recovered toward the average.

2.7 The trading vocabulary used in the results

Finally, six short definitions used throughout Sections 3 and 4. Going long (or "buying") means owning the asset so that you profit if the price rises. A position is a trade that is currently open; you are "in" a position between your entry (the purchase) and your exit (the sale). P&L (profit and loss) is the gain or loss on a trade, expressed here as a percentage of the entry price. The equity curve is a running total of all P&L over time (the line a trader watches to see whether their account is growing.) A drawdown is a fall in the equity curve from its previous peak; the maximum drawdown is the deepest such fall, and it measures the worst pain the strategy would have inflicted. A stop-loss is a pre-set rule that closes a losing trade automatically at a fixed loss (e.g. -4%); the strategy tested here deliberately has none, and Section 5 shows why that matters.

3. Methodology

3.1 Data

Two kinds of data were collected. Daily prices: closing prices of WTI (CL=F) and Brent (BZ=F) front-month futures for every trading day from 2015-01-02 to 2026-07-10 (2896 observations per series) downloaded from Yahoo Finance, which republishes official exchange prices. Weekly supply statistics: US commercial crude inventories and refinery utilisation from the EIA's Weekly Petroleum Status Report. Table 1 defines every column of the resulting dataset.

Table 1. Data dictionary, every column in the workbook, defined.

Column	What it contains	Where it comes from
Date	The calendar trading day (Monday–Friday; exchanges close at weekends).	—
WTI_Close	The last traded price of the front-month WTI futures contract that day, in \$/bbl.	Yahoo Finance, ticker CL=F
Brent_Close	The same, for the Brent futures contract.	Yahoo Finance, ticker BZ=F
Spread	Brent_Close minus WTI_Close (computed by formula).	Derived
US Crude Inventory	Barrels of crude in US commercial storage, millions (weekly; appears on the Friday ending each reporting week).	EIA Weekly Petroleum Status Report
Refinery Util (%)	Share of US refining capacity in operation that week.	EIA Weekly Petroleum Status Report
WTI_20MA	Average of the most recent 20 values of WTI_Close (computed by formula).	Derived
Deviation_%	How far today's price sits from its average: $(WTI_Close - WTI_20MA) \div WTI_20MA \times 100$.	Derived

The period was chosen deliberately to include extreme events: the 2015–16 shale glut, the 2020 COVID-19 collapse (when WTI futures briefly traded below zero, sellers paid buyers to take oil because storage was full), and the 2022 price spike that followed the invasion of Ukraine.

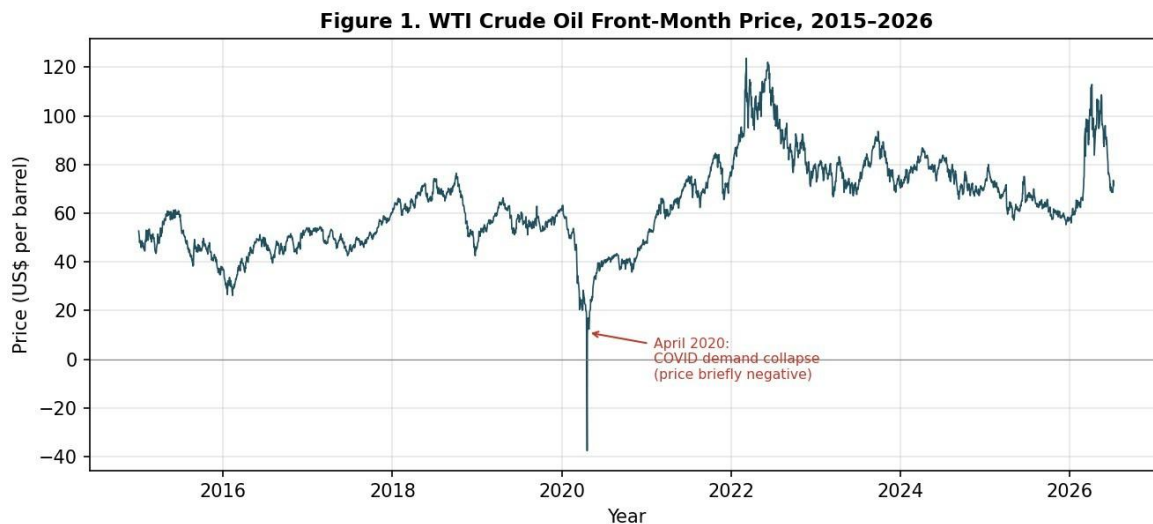


Figure 1. The raw material of this study: 2896 daily closing prices of WTI, 2015–2026. The price ranged from \$-37.63 (briefly negative, April 2020) to \$123.7 (2022). Every result in Section 4 is derived from this series.

3.2 The strategy rule, stated precisely

A backtest requires the rule to be written so precisely that a computer (or a spreadsheet) can execute it with no discretion. The rule tested is:

- **Step 1.** Each day, compute the 20-day moving average (MA20) of the WTI closing price, and the deviation: $(\text{price} - \text{MA20}) \div \text{MA20} \times 100$.
- **Step 2.** If no position is open AND the deviation is below -5% (the price has fallen more than 5% under its one-month average), BUY at that day's closing price. This is the entry.
- **Step 3.** If a position is open AND the deviation rises above 0% (the price has recovered to its average), SELL at that day's closing price. This is the exit.
- **Step 4.** Otherwise, do nothing (HOLD). Only one position may be open at a time, and the position size is the same every trade.

The trade's P&L is $(\text{exit price} - \text{entry price}) \div \text{entry price} \times 100$. Note what the rule does not contain: no forecast, no news reading, no judgement, and no stop-loss. If the price keeps falling after entry, the rule simply waits, however long it takes and however deep the loss becomes. This choice is examined critically in Section 5.

3.3 Implementation in a spreadsheet

The entire test was implemented in Microsoft Excel (workbook "Crude_Oil_Backtest.xlsx", supplied with this report) so that every intermediate number is visible and auditable. The Data sheet holds the prices and computes MA20 and the deviation with ordinary AVERAGE formulas. The Strategy sheet walks through all 2896 days with IF formulas that generate the signal (BUY / SELL / HOLD), track whether a position is open, record entry and exit prices, and compute each trade's P&L plus a running cumulative total. The Results sheet aggregates the trade list into the summary statistics defined below. No step involves anything more advanced than IF, AVERAGE, COUNT and SUM.

3.4 The performance metrics, defined

Results in Section 4 are reported with the industry-standard vocabulary. Each metric is defined here in full:

- **Total trades:** the number of completed round trips (one BUY followed by one SELL).
- **Win rate:** the percentage of trades that ended with a profit. A high win rate feels good but is not sufficient: a strategy can win 90% of the time and still lose money if the rare losses are huge.
- **Average win / average loss:** The mean P&L of profitable trades, and of losing trades, respectively. Comparing them shows whether wins and losses are symmetric.
- **Profit factor:** The sum of all profits divided by the absolute sum of all losses. Above 1.0 the strategy makes more than it loses; 1.5 means every \$1 lost was accompanied by \$1.50 gained.
- **Total P&L:** the sum of all individual trade returns, in percentage points. Because each trade uses the same position size and profits are not reinvested, returns add rather than compound; this keeps the arithmetic transparent for a first study.
- **Maximum drawdown:** The deepest fall of the cumulative P&L from its highest point so far. It answers: “if I had started at the worst possible moment, how much would I have been down?”
- **Buy & hold return:** The return from buying WTI on day one and holding to the end, with no trading at all. This is the benchmark: a strategy is only interesting if it beats doing nothing.

4. Results

4.1 Headline performance

Table 2 reports every metric defined in Section 3.4, computed over the full 11.5-year sample. Each row includes a plain-language interpretation.

Table 2. Backtest results, 2015–2026. “pct-pts” = percentage points.

Metric	Value	Plain-language meaning
Total trades	59	The rule triggered about 5 trades per year.
Winning trades	42	42 of 59 trades made money.
Losing trades	17	17 trades lost money.
Win rate	71.2%	Roughly 7 out of every 10 trades were profitable.
Average win	+5.32%	A typical winner gained about five percent.
Average loss	-8.76%	A typical loser lost about nine percent — bigger than a typical win.
Profit factor	1.5	For every \$1 lost, \$1.50 was won.
Best trade	+11.3%	The single most profitable trade.
Worst trade	-39.8%	One trade — entered just before the 2020 crash — lost almost 40%.
Total P&L (sum of trades)	+74.5 pct-pts	Adding every trade's return gives +74.5 percentage points over 11.5 years.
Maximum drawdown	82.4 pct-pts	At its lowest, the cumulative curve sat 82 points below its earlier peak (driven by 2020).
Buy & hold return	+35.5%	Doing nothing, buy in 2015, hold to 2026, earned 35.5%.
Outperformance	+39 pct-pts	The rule roughly doubled the do-nothing benchmark, before costs.

The headline finding: the rule was profitable, with a high win rate (71.2%) and roughly double the return of buy-and-hold. Figure 5 shows the equity curve, the running sum of all trade returns.

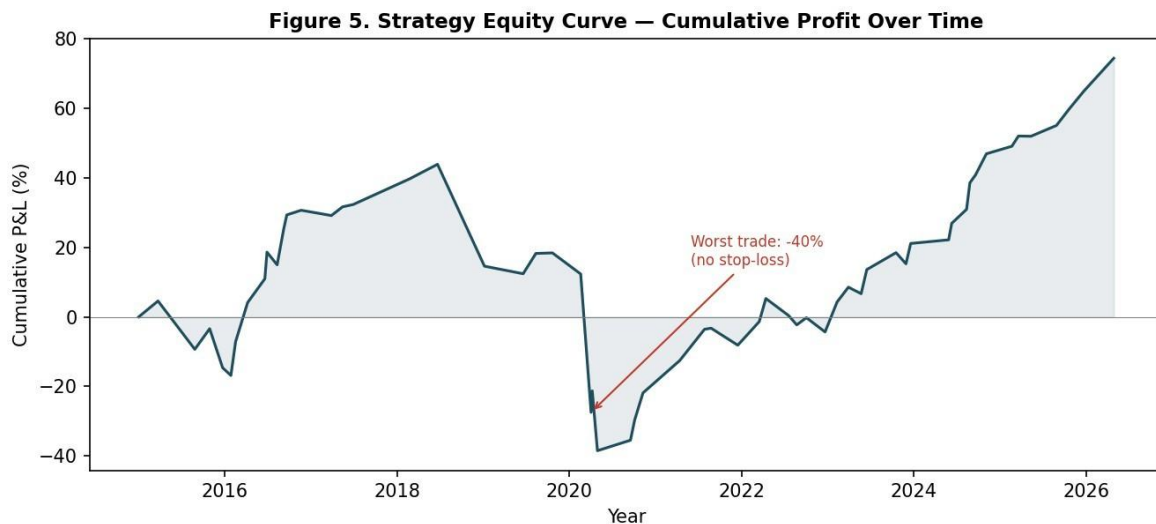


Figure 5. The equity curve: cumulative P&L after each completed trade. The line climbs steadily through 2015–2019, suffers a severe setback in 2020 (annotated), then recovers and reaches new highs. Reading an equity curve: upward slope = the strategy is winning; flat = no trades or break-even; sharp drops = large losing trades.

4.2 The anatomy of wins and losses

Figure 6 ranks all 59 trades from best to worst. The picture is typical of mean-reversion strategies: many moderate winners (green) and a few severe losers (red). The average win (+5.32%) is smaller than the average loss (-8.76%); the strategy remains profitable only because winners outnumber losers roughly 2.5 to 1. This asymmetry is the strategy's signature — and its central risk.

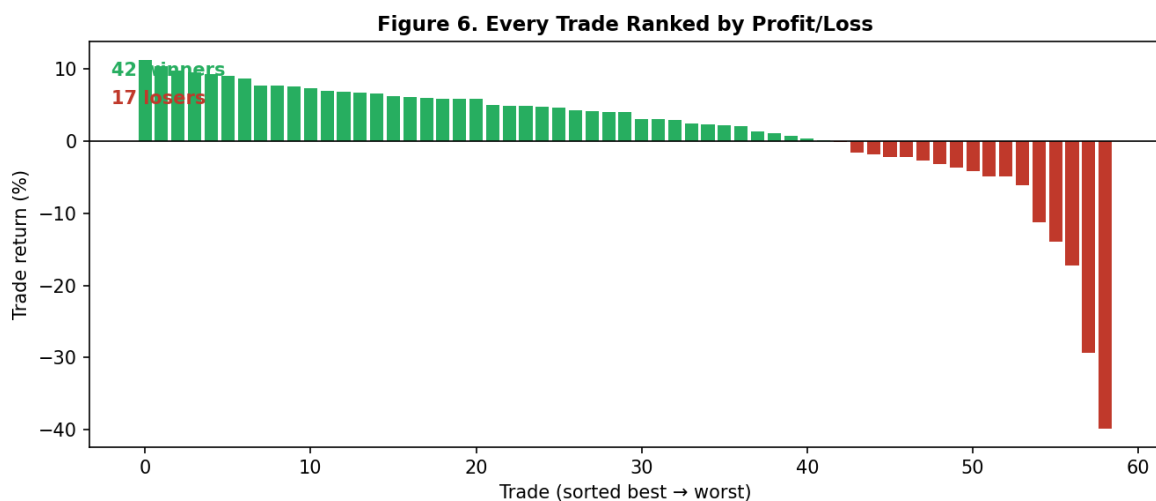


Figure 6. All 59 trades sorted from most profitable (left) to least profitable (right). Most bars are green and modest; the tail of red bars on the right is short but deep the worst bar is the 39.8% trade opened just before the March 2020 collapse.

Table 3 shows five real trades from the record, including the catastrophic 2020 one, so the reader can see exactly what a trade looks like in the data.

Table 3. Five example trades from the full record of 59.

Entry date	Exit date	Bought at (\$)	Sold at (\$)	P&L (%)
2015-03-16	2015-03-25	47.05	49.21	+4.59
2015-07-06	2015-08-28	52.53	45.22	-13.92

Entry date	Exit date	Bought at (\$)	Sold at (\$)	P&L (%)
2015-09-01	2015-10-30	43.98	46.59	+5.93
2020-02-24	2020-06-03	51.43	37.29	-27.50 (COVID crash)
2022-08-01	2022-08-11	93.89	94.34	+0.48

4.3 Strategy versus doing nothing

A strategy must beat its benchmark to justify its existence. Figure 7 overlays the strategy's equity curve on the buy-and-hold return. Two differences stand out. First, the endpoint: +74.5 points versus +35.5%. Second and arguably more important the journey: buy-and-hold spent years underwater (an investor who bought in 2015 was still losing money in 2020), while the strategy was profitable for most of the period because it only holds oil during recoveries and sits in cash otherwise.



Figure 7. Strategy (dark blue) versus buy-and-hold (orange). Buy-and-hold is fully exposed to every crash; the strategy is exposed only while a position is open, which is a minority of the time.

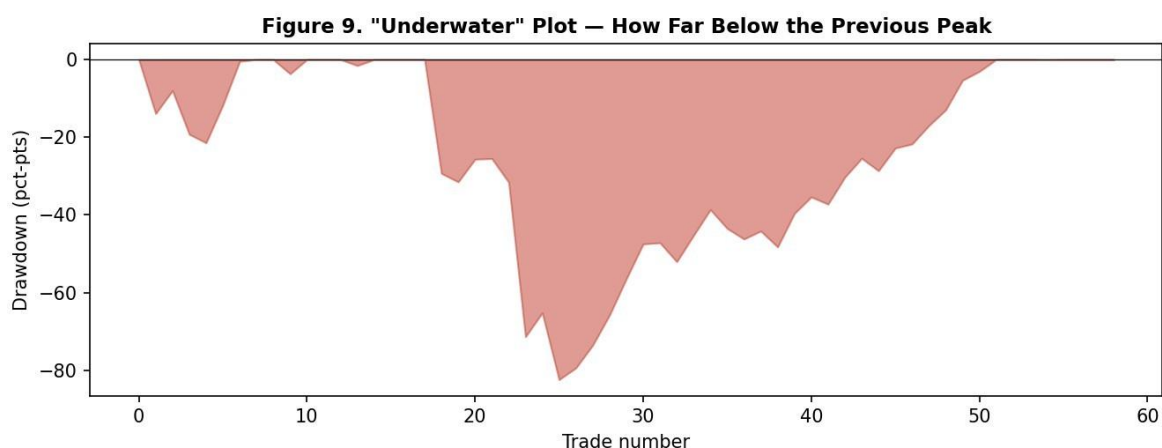


Figure 9. The "underwater" plot: how far the strategy's cumulative P&L sat below its previous peak after each trade. The deepest point (-82 pct-pts region) corresponds to the 2020 sequence of losses.

4.4 The most recent 12 weeks: a live case study

The workbook's Tracking sheet records the last 12 weeks (mid-April to July 2026) with real prices and real EIA statistics, and it illustrates why traders never read a price chart in isolation. Over those weeks, WTI fell from about \$83.85 (and an intra-period peak above \$105 in mid-May) to \$71.41 yet US crude inventories fell too, from 465.7 to 411.4 million barrels, and refinery utilisation ran at an extremely high 95–97%. Falling inventories are normally bullish, so prices falling at the same time is a puzzle worth investigating (candidate explanations include a geopolitical risk premium deflating from the May spike, rising supply from outside the US, or expectations of weaker future demand). Investigating exactly such puzzles is the daily work of a commodity analyst, and Section 6 proposes it as follow-up work.

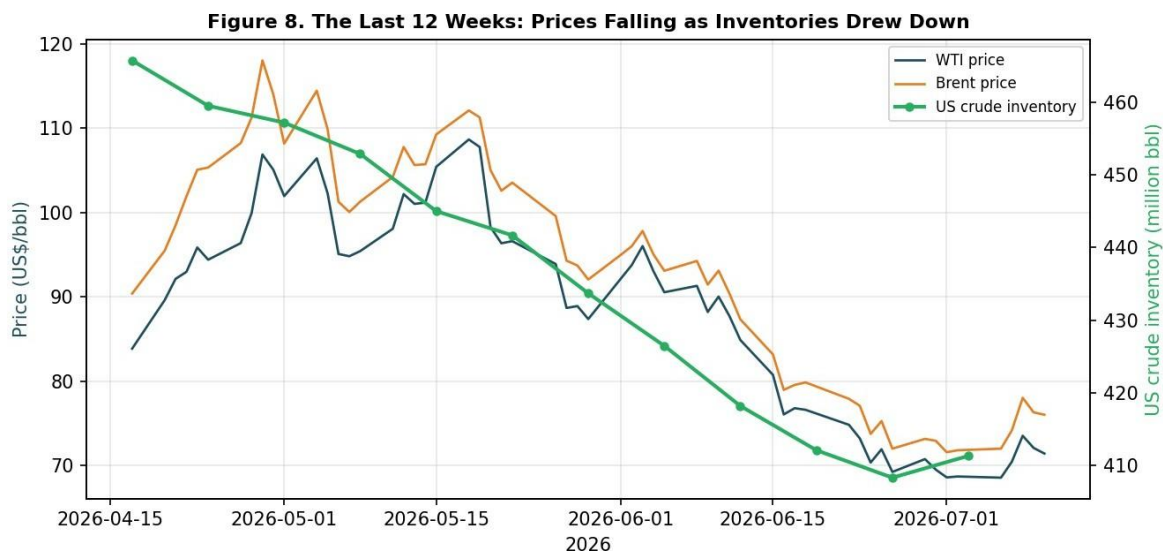


Figure 8. The last 12 weeks of real data from the Tracking sheet. Left axis: WTI (dark blue) and Brent (orange) prices. Right axis: US crude inventories in millions of barrels (green, weekly points). Prices fell roughly \$30 from the May peak even as inventories drew down by more than 50 million barrels, an unusual combination.

5. Discussion

5.1 Why the rule worked

The results are consistent with the physical logic of Section 2.6: sharp oil sell-offs during 2015–2020 were usually followed by recoveries toward the one-month average, because cheap crude stimulates refinery buying and squeezes high-cost producers. The rule mechanically harvests those recoveries. Its edge is behavioural and physical, not informational.

5.2 Why the results overstate real-world profits

- **Transaction costs are ignored.** Every trade pays a broker commission and crosses a bid-ask spread, realistically \$0.05–0.10 per barrel per side. Over 59 trades this shaves several percentage points off the total.
- **Slippage is ignored.** The backtest assumes execution exactly at the closing price. In fast-falling markets (precisely when this rule buys), actual fill prices are often worse, an effect called slippage.
- **Roll costs are ignored.** Futures contracts expire monthly; a continuous position must be “rolled” into the next contract, which costs money when the curve is in contango (Section 2.3). The continuous price series used here hides that cost.
- **Parameter luck is possible.** The -5% / 20-day parameters were not optimised here, but any published rule risks having survived because it happened to fit the past, a bias called overfitting. Robustness checks (testing -3% and -7% , testing on Brent) are proposed in Section 6.

5.3 The missing stop-loss: the strategy's real lesson

The most instructive number in Table 2 is the worst trade: -39.8% . The rule bought in late February 2020 after a 5% dip and then COVID-19 lockdowns erased a third of world oil demand within weeks. With no stop-loss, the rule held through the entire collapse. A simple protective exit at -4% would have capped that loss at one-tenth of its size, at the cost of occasionally being knocked out of trades that would have recovered. The trade-off between “give trades room to recover” and “never take a catastrophic loss” is the central design decision of every real trading system, and this backtest demonstrates it.

5.4 What a backtest can and cannot prove

A clean backtest proves that a rule would have worked in the past, under stated assumptions. It cannot prove the rule will work in the future: market structure changes (US shale, electric vehicles, OPEC policy), and patterns that many traders exploit tend to weaken. Professional practice therefore treats a backtest as a hypothesis test and a risk-measurement exercise rather than as a promise of profit.

6. Conclusions and Further Work

A five-line trading rule, buy WTI when it closes 5% below its 20-day average, sell on recovery, was backtested on 2896 days of real prices spanning 11.5 years. It completed 59 trades, won 71.2% of them, and accumulated +74.5 percentage points against +35.5% for buy-and-hold, before costs. The same test exposed the rule's defining weakness: absent a stop-loss, a single crisis trade lost 39.8%, and the maximum drawdown reached 82.4 points. The study therefore supports a modest, precise claim: crude oil exhibited economically meaningful mean reversion over 2015–2026, and a mechanical rule could capture it provided its risks are understood and managed.

Natural extensions, all implementable in the same workbook:

- 1. add a –4% stop-loss and re-run, comparing win rate, profit factor and maximum drawdown;
- 2. test entry thresholds of –3% and –7% to check the result is not parameter luck;
- 3. repeat the test on Brent (column C of the Data sheet) to check the pattern is not WTI-specific;
- 4. subtract realistic costs (\$0.10/bbl per round trip) from every trade;
- 5. investigate the recent-12-weeks puzzle of Section 4.4 using inventory, OPEC and demand data.

7. Glossary (Alphabetical)

Every specialist term used anywhere in this report, defined in one or two sentences.

Term	Definition
Backtest	Replaying a precisely defined trading rule against historical data to measure how it would have performed.
Backwardation	A futures curve where later delivery months are cheaper than nearer ones; signals tight current supply.
Barrel (bbl)	The standard unit of oil volume: 159 litres (42 US gallons).
Bearish / bullish	Expecting prices to fall (bearish) or rise (bullish).
Benchmark (crude)	A standard reference grade of oil against which other grades are priced. WTI and Brent are the two main ones.
Brent	Light, sweet North Sea crude; the benchmark for most internationally traded oil. Ticker BZ (BZ=F on Yahoo Finance).
Buy & hold	The do-nothing benchmark strategy: buy on the first day of the sample and hold to the last.
Contango	A futures curve where later delivery months are more expensive; signals comfortable supply and rewards storage.
Crude oil	Unrefined petroleum as extracted from the ground, before processing into petrol, diesel, jet fuel, etc.
Deviation (%)	In this report: how far today's price sits from its 20-day average, as a percentage of that average.
Drawdown / maximum drawdown	A fall of the equity curve from its previous peak; the maximum drawdown is the deepest such fall in the sample.
EIA	U.S. Energy Information Administration — the statistics agency of the US Department of Energy; publishes the weekly inventory and refinery data used here.
Entry / exit	The opening purchase of a position, and its closing sale.
Equity curve	The running total of all trade profits and losses over time.
Front-month contract	The futures contract expiring soonest; its price is the widely quoted “price of oil.”
Futures contract	A standardised, exchange-traded agreement to buy or sell 1,000 barrels of crude at a set price on a set future date.
Hedging	Using futures to lock in a price in advance and thereby remove price risk (e.g. an airline fixing its fuel cost).
Light / heavy, sweet / sour	Quality descriptors for crude: light (low density) and sweet (low sulphur) crudes are easier to refine and command higher prices.
Long (going long)	Owning an asset (or futures contract) so that you profit if its price rises.
Mean reversion	The tendency of a price that has moved sharply away from its recent average to return toward it.
Moving average (20-day, MA20)	The arithmetic mean of the last 20 daily closing prices; a smoothed, one-month view of the price.
OPEC	Organization of the Petroleum Exporting Countries, a group of major producers (led by Saudi Arabia) that coordinates output to influence prices.
P&L (profit and loss)	The gain or loss on a trade; here expressed as a percentage of the entry price.
Position	An open trade. “In a position” = currently holding; “flat” or “out” = holding nothing.

Term	Definition
Profit factor	Sum of all profits divided by the absolute sum of all losses; above 1.0 means net profitable.
Refinery utilisation	The percentage of total refining capacity actually operating; a gauge of refiners' demand for crude.
Roll (rolling a contract)	Closing an expiring futures contract and opening the next month's to maintain a continuous position.
Signal	The rule's instruction on a given day: BUY, SELL, or HOLD.
Slippage	The difference between the price a backtest assumes and the worse price actually obtainable in live trading.
Spread (Brent–WTI)	Brent price minus WTI price; a gauge of US-versus-global supply balance.
Stop-loss	A pre-set rule that automatically closes a losing position at a fixed loss to prevent catastrophe.
Ticker (CL=F, BZ=F)	The short code identifying a traded instrument; CL=F and BZ=F are Yahoo Finance's codes for WTI and Brent front-month futures.
Transaction costs	Commissions and bid-ask spreads paid on every trade; excluded from this backtest and discussed in Section 5.
US crude inventory	Barrels of crude oil in US commercial storage (excluding the Strategic Petroleum Reserve); published weekly by the EIA.
Win rate	The percentage of trades that ended profitably.
WTI (West Texas Intermediate)	Light, sweet US crude delivered at Cushing, Oklahoma; the main US price benchmark. Ticker CL (CL=F on Yahoo Finance).

8. Data Sources and Materials

- **1.** Yahoo Finance: daily front-month futures closing prices, tickers CL=F (WTI) and BZ=F (Brent), 2015-01-02 to 2026-07-10. finance.yahoo.com
- **2.** U.S. Energy Information Administration: Weekly Petroleum Status Report: US commercial crude inventories (series WCESTUS1) and refinery utilisation (series WPULEUS3). eia.gov/petroleum
- **3.** Accompanying workbook: Crude_Oil_Backtest.xlsx: sheets Tracking (12-week daily log), Data (full price history and derived columns), Strategy (day-by-day signals and trades), Results (summary statistics and equity chart). All formulas are open and auditable.
- **4.** Suggested introductory reading: J. Hull, Options, Futures, and Other Derivatives (chapters 1–5); O. Johnson, 40 Classic Crude Oil Trades; A. Imsirovic, Trading and Price Discovery for Crude Oils.